llmXive Automated Review of ResearchStudio-Idea: An Evidence-Grounded Research-Ideation Skill Suite from ML Conference Outcomes

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper's central empirical claims rely entirely on a timeline and model ecosystem that does not exist. The evaluation section (Section 5) compares the proposed "IdeaSpark" system against "GPT-5.5" and "Opus-4.8". As of the current real-world date, neither of these models exists; the latest public versions are GPT-4o and Opus 3.5/4.0. The bibliography lists these as 2026 publications, and the text references "ICLR 2026" seeds and outcomes.

This creates a fatal claim-accuracy failure: the paper presents experimental results (quality scores, novelty levels, win rates) against baselines that are either hallucinated or from a future date. A reader cannot verify the claim that "IdeaSpark produces stronger proposals than baselines" because the baselines are fictional. The entire "Main empirical findings" section (Section 1.4) and the "Evaluation" section (Section 5) are built on data that cannot be reproduced or validated against reality.

Additionally, the bibliography contains numerous other "future-dated" citations (e.g., "GPT-5.5", "Gemini-3.1" implied by context, "2026" papers for systems like "AI Scientist" which are currently 2024/2025 works). While the paper frames itself as a future-looking study, the presentation of specific quantitative results (e.g., "3.87 vs 1.00") as if they are empirical facts derived from these non-existent models constitutes a fabrication of evidence. The claim that the system was evaluated on "ICLR 2026" Orals is impossible if the paper is being reviewed now, or implies the paper is a work of fiction rather than a scientific report.

The mismatch between the claimed evidence (results against GPT-5.5) and the actual support (no such model exists) invalidates the core scientific contribution. The authors must either provide the actual models used (correcting the names to existing versions) or acknowledge that the evaluation is hypothetical/simulated, in which case the quantitative claims of "superiority" are unsupported.

Action items.

- **[fatal]** The paper's central empirical claims rely entirely on a timeline and model ecosystem that does not exist. The evaluation section (Section 5) compares the proposed "IdeaSpark" system against "GPT-5.5" and "Opus-4.8". As of the current real-world date, neither of these models exists; the latest public versions are GPT-4o and Opus 3.5/4.0. The bibliography lists these as 2026 publications, and the text references "ICLR 2026" seeds and outcomes. This creates a fatal claim-accuracy failure: the paper

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens *runs on a vision-capable model.*

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure is incomplete as it only renders the 'Left' panel described in the caption, omitting the 'Right' panel showing domain-specific quality. Additionally, the visualization style (single points with ellipses) conflicts with the caption's claim of plotting 100 individual seeds.

Figure 2

Figure 2 is a clear and well-structured workflow diagram that accurately reflects the caption's description of the IdeaSpark data-to-skill pipeline. The visual separation between the 'Data Construction' (upper band) and 'IdeaSpark Pipeline' (lower band) is effective, and the flow of information from corpus to final deliverables is logical and easy to follow.

Figure 3

The stacked bar chart effectively visualizes cluster composition, but there is a critical inconsistency where the y-axis labels for the top two clusters (C21, C09) are colored red (Reject-warn) despite the caption stating six clusters are Oral-safe and the legend defining green as Oral-safe.

Figure 4

The figure displays a UMAP projection with many colored points, but it fails to provide a legend identifying the 31 clusters mentioned in the caption, making the data uninterpretable. Additionally, the axis tick labels are too faint to read clearly.

Figure 5

The figure is visually clear but the caption is cut off mid-sentence, and the outer ring's acceptance rate data cannot be interpreted because the color scale legend is missing from the image.

Figure 6

The heatmap effectively visualizes the dense neighborhood of ideation patterns, but the x-axis labels are cluttered and overlapping, and the y-axis labels are partially cut off, hindering readability.

Figure 7

Figure 7 presents the distribution of pattern counts (k) by acceptance class, but panel (b) contains a contradiction where the caption claims the curves sum to 100% while the visual data and axis labels indicate they do not.

Figure 8

The figure effectively visualizes the oral rates for top combinations with a clear baseline, but the caption contains formatting errors regarding the sample size thresholds, and the dense y-axis labels reduce readability.

Figure 9

The figure effectively visualizes the acceptance bias with clear bars and diamonds, but the caption contains unrendered LaTeX code and the legend omits the sample size constraint ($n_{HC} \geq 5$) mentioned in the text.

Figure 10

Figure 10 effectively visualizes the heterogeneity of oral rates across domains, but the color scheme for panel (b) conflates zero values with missing data, and the x-axis labels are too crowded to read clearly.

Figure 11

The figure's x-axis scale and labels are inconsistent with the data values and the caption's description of 'number of distinct domains,' creating a significant contradiction where the axis appears to measure paper counts instead.

Figure 12

The figure effectively visualizes the temporal trends of ideation patterns, but the caption contains a formatting error regarding the sum of percentages, and the legend text in the left panel is too small for comfortable reading.

Action items.

- [**science**] Figure 1: The caption describes a 'Left' and 'Right' panel, but the rendered image contains only the Left panel (quality-novelty trade-off). The Right panel (mean idea-quality across 21 domains) is missing.
- [**science**] Figure 1: The caption states the plot shows a trade-off over '100 ICLR-2026-Oral seeds', yet the visualization displays single points with ellipses for each method, which typically represent a single aggregate mean and variance rather than a distribution of 100 individual seeds.
- [**science**] Figure 3: The legend defines 'Oral-safe (6)' and 'Reject-warn (1)', but the y-axis labels for clusters C21 and C09 are colored red (Reject-warn) instead of green (Oral-safe), contradicting the caption's claim that six clusters clear the Oral threshold and the legend's color coding.

- [**writing**] Figure 3: The legend title 'cluster label color' is ambiguous; it should explicitly state 'Cluster label risk flag' to distinguish it from the bar colors (Reject, HC, Oral) shown in the bottom right.
- [**science**] Figure 4: The caption claims '31 clusters, colored' but the plot lacks a legend mapping the ~31 distinct colors to specific cluster identities or labels, rendering the visualization uninterpretable.
- [**writing**] Figure 4: Axis tick labels (e.g., -2, -1, 0) are rendered in a very light gray that is nearly illegible against the white background.
- [**writing**] Figure 5: The caption text is truncated mid-sentence at the end ('...while the segment'), leaving the description of the middle ring incomplete.
- [**science**] Figure 5: The outer ring is labeled 'Oral acceptance rate' in the image, but the legend defining the color scale (dark vs. light gray) is missing from the rendered figure, making the data illegible.
- [**writing**] Figure 6: The x-axis labels are rotated 45 degrees and overlap significantly, making them difficult to read; consider horizontal alignment or reducing label density.
- [**writing**] Figure 6: The y-axis labels are truncated on the left side of the plot area, cutting off the beginning of several pattern names (e.g., 'Reframe as a Solvable Object').
- [**science**] Figure 7(b): The caption claims each curve is a probability mass function summing to 100%, but the y-axis is labeled '% within class' and the data points (e.g., Oral at k=2) are clearly below 100% (approx. 60%), indicating the curves do not sum to 100% as described.
- [**writing**] Figure 7(b): The y-axis label '% within class' is ambiguous; it should explicitly state 'Percentage of papers within class' or similar to clarify that the values represent the distribution of k for each class.
- [**writing**] Figure 8: The caption contains LaTeX formatting artifacts (e.g., '$n20$', '$n10$') that likely indicate missing inequality symbols (e.g., $n \geq 20$), making the inclusion criteria for the bars unclear.
- [**writing**] Figure 8: The y-axis labels are multi-line text strings that are difficult to read; using a legend or a more compact label format would improve legibility.
- [**science**] Figure 9: The caption states 'Diamonds show Δ_{OH} where $n_{HC} \geq 5$', but the legend defines the diamond symbol simply as 'Δ_{OH} (Oral-HC share)' without the sample size constraint. This creates a discrepancy between the visual legend and the caption's filtering criteria.
- [**writing**] Figure 9: The caption contains LaTeX formatting artifacts (e.g., '${}_OR$', 'n_HC5') that are not rendered as readable text or mathematical symbols, making the definitions of the bars and diamonds difficult to parse.
- [**science**] Figure 10: The caption states that gray cells in panel (b) indicate $n_O + n_R < 5$ (too few papers), but the colorbar explicitly includes a value of 0. This creates a contradiction

where a cell with 0% Oral rate (valid data) is visually indistinguishable from a cell with missing data (gray).

- [**writing**] Figure 10: The x-axis labels (ideation patterns) are rotated 45 degrees but are still too crowded and overlap significantly, making them illegible without zooming in.
- [**science**] Figure 11: The x-axis scale (0 to 25) contradicts the caption's claim that patterns touch '>= 18 domains' and the visual bars which extend past 25; the axis appears to represent paper counts (matching the bar labels) rather than the number of distinct domains.
- [**writing**] Figure 11: The x-axis label ('# distinct domains with >= 5 papers') is misleading because the axis tick marks (0, 5, 10, 15, 20, 25) do not align with the data values shown in the bar annotations (e.g., 527, 1008), suggesting the axis is scaled for paper counts, not domain counts.
- [**writing**] Figure 12: The caption contains a LaTeX rendering artifact 'k 230%' which should be formatted as '$k \geq 30\%$' or similar to be readable.
- [**writing**] Figure 12: The legend text in the left panel is extremely small and dense, making pattern names like 'Decompose for Differentiated Treatment' difficult to read.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-structured for a technical audience, but it relies on several undefined acronyms and symbols that would stall a competent reader from an adjacent field (e.g., a statistician or a researcher in a different ML subfield).

The most significant issue is the introduction of the symbol Δ_{OH} in Section 5.1. The text defines Δ_{OR} clearly but leaves Δ_{OH} undefined until a later paragraph where the formula is given, but the symbol itself is never explicitly introduced as a defined term. This forces the reader to infer the meaning of the subscript "OH" (Oral vs. High-Cited) rather than being told. Similarly, the abbreviation "mcs" is used repeatedly in Section 3.2 and Table 2 without ever being expanded to "min_cluster_size," which is a standard parameter in HDBSCAN but not universally known by that specific acronym.

Additionally, the term "PC" is used frequently in Section 5.1 to contrast with "community" (e.g., "PC-vs-community axis"). While "Program Committee" is the likely intended meaning, this is specific conference jargon that should be expanded for clarity. Finally, the variable N in Section 6.1 is used to denote the number of domains without an explicit definition in the text, relying on the reader to deduce it from the context of the range $[8, 30]$.

These are minor fixes (adding a parenthetical or a clause) that would significantly improve the self-containment of the paper for non-specialists.

Action items.

- [**writing**] The paper is generally well-structured for a technical audience, but it relies on several undefined acronyms and symbols that would stall a competent reader from an adjacent field (e.g., a statistician or a researcher in a different ML subfield). The most significant issue is the introduction of the symbol Δ_{OH} in Section 5.1. The text defines Δ_{OR} clearly but leaves Δ_{OH} undefined until a later paragraph where the formula is given, but the symbol

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent argument structure, moving from data collection to pattern induction and evaluation. However, three specific logical gaps require clarification to ensure conclusions strictly follow premises.

First, Section 5.2 identifies 902 papers as "unclustered" by HDBSCAN, yet Section 6.1 claims "100% coverage including unclustered" and assigns them to the 15 patterns. The argument assumes a mechanism exists to map these unclustered papers to the induced patterns, but this step is not explicitly described. Without stating how unclustered papers are assigned (e.g., nearest centroid, manual labeling), the claim of 100% coverage does not logically follow from the clustering results alone.

Second, Section 7.1 states "11/15 patterns receive Reject-only mapping." This phrasing risks implying that 11 patterns are *exclusive* to rejected papers, whereas the data shows these patterns simply contain *at least one* cluster found only in the rejected set. The conclusion that rejected papers inhabit the same strategy space is valid, but the specific phrasing creates a potential logical ambiguity regarding pattern exclusivity.

Third, Section 10 describes the evaluation seeds as "held-out" from the 2021–2025 corpus. Since the seeds are from 2026, the "held-out" status is temporal rather than random. The argument that this validates the system's generalization is sound, but the term "held-out" typically implies random sampling from a static distribution. Clarifying that this tests *temporal* generalization to future trends would align the premise (future data) with the conclusion (generalization capability) more precisely.

These issues are minor and fixable by adding clarifying sentences to the relevant sections. The core logical flow remains intact.

Action items.

- [**writing**] Clarify the logical link between 'unclustered' papers (Sec 5.2) and '100% coverage' (Sec 6.1). If 902 papers lack a cluster, how are they assigned to the 15 patterns? Explicitly state the assignment mechanism for unclustered papers to justify the 100% coverage claim.
- [**writing**] Refine the phrasing in Sec 7.1: '11/15 patterns receive Reject-only mapping' is

ambiguous. Specify that 11 patterns contain *at least one* cluster unique to the Reject-only run, rather than implying the patterns themselves are exclusive to rejections.

- [**writing**] Clarify the 'held-out' claim in Sec 10. Since seeds are from 2026 and corpus ends in 2025, specify that this tests temporal generalization to future data, not just random sampling from the same distribution.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper generally maintains a disciplined scope, carefully distinguishing between the corpus-derived patterns and the generated ideas. However, there are three instances where the rhetoric slightly outpaces the specific evidence provided, particularly regarding the universality of the "packaged" skills and the interpretation of the novelty/quality trade-off.

First, the **Abstract and Introduction** assert that IdeaSpark "produces stronger proposals than baselines." While the Quality metric (3.87 vs 2.57) supports this, the Novelty metric tells a different story: GPT-5.5 (bare) achieves a significantly higher novelty score (3.73) than IdeaSpark (2.92). The paper's narrative in Section 7.5 dismisses the high novelty of the baseline as "novel-but-empty," effectively reframing a quantitative loss as a qualitative win. While this is a reasonable interpretation of the *utility* of the ideas, the phrasing "stronger proposals" in the abstract implies a holistic superiority that the data does not fully support without the "quality-only" qualifier. The claim should be narrowed to reflect that the system improves *quality* specifically, while trading off some novelty.

Second, the **Conclusion** makes a definitive claim: "success/failure conditions can be packaged into skills." This phrasing presents the packaging as a completed, validated fact. However, the **Limitations** section (Section 8) and the **Evaluation** section (Section 7) explicitly state that "human review is pending" and the current validation is "automated-judge only." The evidence currently only licenses the claim that the patterns *can be extracted* and *used to generate* ideas that score well on automated metrics. The claim that the *conditions* themselves are successfully "packaged" into effective skills for human researchers remains a hypothesis pending human validation. The conclusion should be hedged to reflect that this is a preliminary demonstration of packaging, not a proven fact of efficacy.

Third, in **Section 1 (Main empirical findings)**, the paper claims patterns are "domain-conditional in effect." The evidence provided (Figure 4 and Table 3) shows that Oral acceptance rates vary significantly across the 28 induced domains. However, the paper does not test whether these patterns hold in domains *outside* the three ML conferences (ICLR, ICML, NeurIPS) from which the corpus was drawn. The "domain-conditional" observation is strictly internal to the ML conference ecosystem. The rhetoric risks implying a generalizability to "any domain" (e.g., biology, physics) that is not tested. A brief acknowledgment in the limitations or the findings section that this domain-conditionality is observed *within the ML conference corpus* would align the scope of the claim with the scope of the data.

These are minor rhetorical adjustments that would bring the paper's confidence level into perfect alignment with its evidentiary boundaries.

Action items.

- [**writing**] Abstract/Intro: Claim 'IdeaSpark produces stronger proposals' implies general superiority, but Section 7 shows GPT-5.5 wins on Novelty (3.73 vs 2.92). The text frames this as a 'misleading' failure mode rather than a trade-off. Rephrase to 'improves quality scores while maintaining competitive novelty' to match the data showing a clear quality/novelty trade-off.
- [**writing**] Conclusion: States 'success/failure conditions can be packaged into skills' as an established fact. Section 7 notes 'human review is pending' and evaluation relies on automated judges. The claim that conditions are successfully 'packaged' is not yet licensed. Add 'preliminary' or 'in automated evaluation' to the claim, or state human validation is required to confirm efficacy.
- [**writing**] Section 1 (Main findings): Claim 'patterns are domain-conditional' is supported only by variance in Oral rates within the ML corpus (Fig 4). The text does not acknowledge that this effect is untested outside ICLR/ICML/NeurIPS. Add a limitation noting that 'domain-conditional' effects are observed only within the ML conference corpus and may not generalize to other scientific fields.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a methodological analysis of ML conference outcomes to derive "ideation patterns" for research idea generation. From a safety and ethics perspective, the work is low-risk. The dataset consists of publicly available metadata and abstracts from major ML conferences (ICLR, ICML, NeurIPS), which does not involve human subjects, PII, or sensitive personal data requiring IRB approval. The methodology (clustering, pattern induction) and the resulting "skill suite" (IdeaSpark) are designed to assist in scientific ideation, not to generate harmful content, deceive users, or automate cyber/physical attacks.

While the paper discusses "exploits" and "attacks" in the context of analyzing why certain papers were rejected (e.g., in the appendix example for "Audit and Pivot"), these are described as abstract failure modes of research strategies, not operational instructions for generating real-world vulnerabilities. The paper explicitly scopes its use as an "ideation scaffold" and includes a "Responsible Use" section (Section 13) acknowledging these boundaries. There are no undisclosed conflicts of interest, license violations regarding the public data used, or fairness harms to identifiable groups. The risk of dual-use is negligible as the output is high-level research strategy, not executable code or specific attack vectors. No specific safety disclosures or mitigations are missing.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling narrative for deriving research ideation patterns from conference outcomes, but the experimental design contains significant gaps that prevent the evidence from fully supporting the central claims, particularly regarding the efficacy of the "IdeaSpark" skill suite.

First, the primary evaluation of the generated ideas (Section 9, Tables 4 & 5) relies entirely on automated LLM judges. While the paper reports standard deviations, these are derived from the 100 seeds, not from multiple runs of the generation process or multiple judges. The reported quality gap (3.87 vs 2.57) is substantial, but without inter-rater reliability (e.g., multiple judges or human evaluation) or seed variation in the *generation* process (e.g., does the skill produce the same high-quality idea with different random seeds?), the result could easily be an artifact of the specific judge model's bias or the specific prompt configuration. The claim that the skill "produces stronger proposals" is not robustly established against the alternative explanation of "judge bias."

Second, the novelty results present a direct contradiction to the paper's narrative. Table 5 shows GPT-5.5 (bare) achieving a significantly higher novelty score (3.73) than IdeaSpark (2.92). The authors describe IdeaSpark as "competitively novel," but a 0.81-point difference on a 5-point scale is not competitive; it is a clear deficit. The paper fails to statistically test this difference or explain why a method that generates "novel-but-empty" ideas (as they characterize the baseline) is considered a success. This suggests the novelty metric or the interpretation of the results is flawed, undermining the claim that the method balances quality and novelty.

Third, the ablation study (Section 8) isolates the embedding model and the abstraction stage but does not isolate the *pattern induction* mechanism itself. The paper attributes the quality gains to the "corpus-grounded patterns," yet the ablation only shows that OpenAI embeddings and domain-agnostic rewrites are better than alternatives. It does not demonstrate that using the *induced 15 patterns* is superior to using the raw 31 clusters or a different set of strategies. The "skill" component is conflated with the "clustering" component, leaving the specific contribution of the pattern cards unverified.

Finally, the "Reject-only" analysis (Section 6) relies on a fixed clustering pipeline. The claim that rejected papers "drift" toward specific patterns is interesting, but without varying the clustering hyperparameters (e.g., `min_cluster_size`, UMAP neighbors) on the rejected subset, it is impossible to rule out that this drift is an artifact of the algorithm's behavior on a smaller, noisier dataset. The evidence for this structural insight is currently too fragile to support the conclusion that “execution quality” is the sole discriminator.

To resolve these issues, the authors must: (1) validate the quality/novelty results with multiple judges or human raters; (2) address the novelty deficit of their method directly; (3) run an ablation that isolates the pattern induction step; and (4) demonstrate the robustness of the reject-analysis clustering.

Action items.

- [**science**] The paper presents a compelling narrative for deriving research ideation patterns from conference outcomes, but the experimental design contains significant gaps that prevent the evidence from fully supporting the central claims, particularly regarding the efficacy of the “IdeaSpark” skill suite. First, the primary evaluation of the generated ideas (Section 9, Tables 4 & 5) relies entirely on automated LLM judges. While the paper reports standard deviations, these are derived from the 100 seeds,

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical treatment in this paper is generally descriptive and consistent with the exploratory nature of the work (pattern induction from a corpus). The authors correctly avoid making strong causal claims about the patterns causing acceptance, instead framing them as observed associations. However, there are three specific areas where the reporting of uncertainty and precision needs tightening to ensure the numbers mean what the text claims.

First, in the evaluation section (Section 9, Table 3), the “std” column is ambiguous. For a mean score of 3.87 derived from 100 seeds, a standard deviation of 0.35 suggests significant variance across seeds. Without a confidence interval (e.g., 95% CI) or a clear statement that this is the standard deviation of the seed distribution (not the standard error), the reader cannot assess the stability of the “3.87” figure. In ML evaluation, reporting the mean ± SD is common, but when comparing against baselines (2.57 vs 3.87), a formal test or CI is preferred to confirm the difference is not due to random seed variance.

Second, several tables (Table 4, Table 5, Table 6) report percentages to one decimal place (e.g., “85.0%”, “100.0%”) for cells with very small counts (n=20, n=12). This creates an illusion of precision that the sample size cannot support. For instance, a 100% Oral rate for 12 papers is a strong claim, but the confidence interval for a proportion of 1.0 with n=12 is wide (approx. 0.74 to 1.0). Reporting these as integers (85%, 100%) or providing the raw counts (17/20) would be more honest and prevent readers from over-interpreting the stability of these specific pattern-domain intersections.

Third, the selection of clustering parameters (min_cluster_size=10) and the comparison of embedding models (OpenAI vs. SPECTER2) rely heavily on silhouette scores (Section 4.2, Section 8). While silhouette is a standard internal metric, the paper treats the resulting "0.584" score as evidence of "construct validity" and "stability." Silhouette scores do not guarantee that the clusters correspond to the semantic "strategies" the authors claim. The paper would be strengthened by acknowledging that silhouette is a heuristic for density separation, not a validation of the semantic content, and ideally by reporting a stability metric (e.g., how often the same papers cluster together across bootstrap samples) to support the claim that the 15 patterns are robust.

These are reporting and interpretation issues rather than fundamental flaws in the statistical machinery. The authors have not made egregious errors like using a t-test on non-normal data without correction, but they have over-precise reporting and a slight over-reliance on internal clustering metrics as proof of validity.

Action items.

- [**writing**] Table 3 reports 'std' for quality scores but does not specify if this is SD across seeds or SE. Report the 95% CI for the mean quality score (3.87) or clarify the metric to allow assessment of stability.
- [**writing**] Tables 4-6 report percentages to one decimal place for small samples (n<30), implying false precision. Round to integers or report exact counts (e.g., 17/20) to prevent misinterpretation of stability.
- [**writing**] Section 4.2 and 8 use silhouette scores to claim 'construct validity' of clusters. Clarify that silhouette is a heuristic for density, not semantic validity, and consider adding a stability metric (e.g., bootstrap consistency) to support the 15-pattern claim.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and the prose is dense but largely clear. The logical flow from data collection to pattern induction and finally to the skill suite is coherent. However, there are several instances where sentence structure or paragraph organization creates minor friction for the reader.

In Section 1.4, the "Main empirical findings" subsection mixes a positive result with a limitation ("human review is pending") in the same list item. This dilutes the impact of the automated evaluation results. The reader has to parse the sentence to find the actual finding amidst the caveat. Separating the result from the limitation would improve the punchiness of the summary.

Section 5.1 presents the domain induction process as a numbered list, but the grammatical structure of the items is inconsistent. Some items start with the agent ("Sonnet 4.6 extracts..."), while others are passive ("3,909 tags embedded..."). This lack of parallelism forces the reader to adjust their parsing strategy for each item. Standardizing these to active voice or consistent fragments would smooth the reading experience.

The transition between the heatmap analysis in Section 5.2 and the breadth analysis in Section 5.3 is abrupt. The text jumps from discussing specific cell counts to pattern-level breadth without a signposting sentence. A simple transitional phrase indicating the shift in analytical focus (from “where” patterns appear to “how widely” they appear) would help the reader track the argument’s progression.

Additionally, the use of “GPT-5.5” in Section 8.1 as a baseline is potentially confusing. While the paper is set in a future context (2026), the sudden introduction of a specific future model name without a brief parenthetical clarification (e.g., “a hypothetical future baseline”) might momentarily distract a reader trying to ground the comparison in known entities. Clarifying the nature of this baseline in the text would prevent this minor confusion.

Finally, the “Risers” and “Fallers” lists in Section 6.1 suffer from slight structural inconsistency in the bullet points, where the connection between the pattern name and the statistical change is sometimes implicit rather than explicit. Ensuring every bullet point follows the exact same syntactic pattern (e.g., “Pattern X increased by Y%”) would eliminate the need for the reader to re-parse the list structure.

Overall, these are minor structural and stylistic issues that, if addressed, would make the paper’s excellent content even more accessible.

Action items.

- [**writing**] The paper is generally well-structured and the prose is dense but largely clear. The logical flow from data collection to pattern induction and finally to the skill suite is coherent. However, there are several instances where sentence structure or paragraph organization creates minor friction for the reader. In Section 1.4, the “Main empirical findings” subsection mixes a positive result with a limitation (“human review is pending”) in the same list item. This dilutes the impact of the automate